HRP-503-TEMPLATE PROTOCOL-Surveys Interviews Questionnaires Focus Groups

1) Protocol Title

a) Title:
Information and Decision Making

b) Protocol Version Date:
March 31, 2022

2) Objectives

a) Describe the purpose, specific aims, or objectives:
The current standard for eliciting beliefs in economics experiments is using incentive compatible elicitation mechanisms. Theoretically, these mechanisms have the property that truth-telling is the dominant strategy. However, recent evidence has shown that some theoretically incentive compatible mechanisms may not translate into the laboratory, and that revealing monetary incentives may harm truthful reporting. In this project we investigate the impact of incentive information on truth telling in the commonly used BDM belief elicitation mechanism.

b) State the hypotheses to be tested:
Hypothesis 1: Revealing the full quantitative incentive information of BDM mechanism may lead to decreased levels of truthful reporting when compared to not revealing incentive information.
Hypothesis 2: Additional training rounds before the belief elicitation task might improve subject understanding and truth telling, holding constant the reveal of quantitative incentive information.
Hypothesis 3: Belief elicitation using introspection with a flat payment may have higher levels of truth telling as compared to BDM mechanism with full quantitative incentive information.

3) Background

a) Describe the relevant prior experience and gaps in current knowledge:
Eliciting beliefs of subjects is of great interest to experimental economics. Being able to observe people's beliefs is important for making inference of decisions and actions in games, for example, and it is central to studying the formation and updating of beliefs. The current state of the art belief elicitation mechanisms include the Quadratic Scoring Rule (QSR) that is based on Brier (1950), the Binarized Scoring Rule (BSR) developed by Hossain and Okui (2013), as well as the BDM mechanism (Becker et al., 1964; Karni, 2009).
The common property of these mechanisms is that they are incentive compatible, meaning the subject's dominant strategy is to always report their belief truthfully. However, there recently there is evidence that theoretical incentive compatibility may not fully translate into behavioral incentive compatibility in the lab, as demonstrated by an experiment using the BSR mechanism by Danz et al. (2020). To date, there has not been any study that directly investigates the behavioral incentive compatibility of the BDM elicitation mechanism in the lab. This project will provide the first experimental

evidence on the behavioral incentive compatibility of BDM belief elicitation mechanism, and how revealing quantitative incentive information of the BDM mechanism may impact truth telling.

b) Describe any relevant preliminary data:
N/A

c) Describe how this research will add to existing knowledge or how the outcomes of this project will be used:
BDM belief elicitation is a widely used elicitation mechanism in economics experiments. A number of recent experiments that study belief updating utilize the BDM elicitation procedure, for example, Mobius et al. (2014), Buser et al. (2018), Coutts (2019), and Coffman et al. (2021). The results of this project will improve our understanding of the real-world performance of the BDM mechanism, and inform experimenters about necessary considerations in its practical implementation.

References

Becker, G. M., M. H. DeGroot, and J. Marschak (1964). Measuring utility by a single-response sequential method.Behavioral Science 9(3), 226–232.

Brier, G. W. (1950). Verification of Forecasts Expressed in Terms of Probability.MonthlyWeather Review 78(1), 1–8.

Buser, T., L. Gerhards, and J. van der Weele (2018). Responsiveness to feedback as apersonal trait.Journal of Risk and Uncertainty 56(2), 165–192.

Coffman, K. B., P. U. Araya, and B. Zafar (2021). A (Dynamic) Investigation of Stereo-types, Belief-Updating, and Behavior.

Coutts, A. (2019). Good news and bad news are still news: experimental evidence onbelief updating.Experimental Economics 22(2), 369–395.

Danz, D., L. Vesterlund, and A. J. Wilson (2020). Belief elicitation: Limiting truthtelling with information on incentives.

Hossain, T. and R. Okui (2013). The binarized scoring rule.Review of Economic Stud-ies 80(3), 984–1001.

Karni, E. (2009). A Mechanism for Eliciting Probabilities.Econometrica 77(2), 603–606.

Mobius, M. M., M. Niederle, P. Niehaus, and T. S. Rosenblat (2014). Managing Self-Confidence.

4) Enrollment Numbers

a) Total number of subjects to be enrolled in this study:
420

b) If this is a reliance, list the number of subjects to be enrolled at each site, by site. (UC Davis: #, Relying Site 1: #, Relying Site 2: #)

N/A

c) Provide a rationale (e.g. statistical justification, power analysis) for the number of subjects to be enrolled.
This study has 4 treatments and each treatment will enroll an equal number of subjects. The main effect of interest is the difference in the proportion of truthful belief reports between the Full Incentive Information treatment and the No Incentive Information treatment. Assuming an effect size of 0.5, a Type I error rate of 0.05, and a power of 0.95, a power analysis using the G-Power software produced a required sample size of 105 for each treatment. Therefore, we expect to enroll a total of 420 subjects between all 4 treatments.

5) Inclusion and Exclusion Criteria

a) Inclusion Criteria:

b) Exclusion Criteria:
Non-US based Prolific and Amazon Mechanical Turk accounts, and UC Davis students who do not have Venmo or PayPal accounts.

c) Age Range:
Above 18 years old.

d) If applicable, describe the screening procedures that you will use to collect data:

6) Procedures Involved

☒ Surveys – Upload all surveys you will use in this study to IRBNet.

☐ Interviews – Upload an interview script with the questions that will be asked during the interview to IRBNet.

☐ Focus groups – Upload a summary of the questions and issues that will be discussed during the focus sessions to IRBNet.

☐ Observation

a) Describe the behavior you will be observing and the setting of the observation:

b) Describe what you will be collecting or documenting for the research:

☐ Other

a) Describe any other data collection or research procedures you will be conducting:

7) Study Timelines

a) Duration of an individual subject's participation in the study:
Around 30 minutes.

b) Estimated timeline to enroll all study subjects:
Subject enrollment is expected to start after IRB approval, and continue until December 2024.

8) Recordings

This research involves:

☐ Audio recordings

☐ Photographs

☐ Video recordings with audio

☐ Video recordings without audio

☒ None of the above

9) Data Management and Confidentiality

Before completing this section, see Privacy and Confidentiality and HIPAA Guidance.

a) List any identifiers that will be collected during the course of this study (e.g., name, medical record number, date of birth, video recordings, etc.):
Amazon Mechanical Turk worker ID, Prolific worker ID, Venmo account name, Paypal account name.

b) If any identifiers will be stored, how long will they be kept?
Identifiers will be destroyed as soon as the subject completes the study and payment is issued. No identifiers will be stored.

c) For data that is coded with a linking key, at what point will the linking key be destroyed?
N/A

d) For any recordings, at what point will the recordings be destroyed?
N/A

NOTES ABOUT USE OF RECORDS

UC Davis Medical Records: UC Davis Health Electronic Health Record (EMR/EPIC) also contains the clinical data for Marshall Medical Center (MMC). MMC patient data cannot be accessed for research purposes. Researchers must take the necessary steps to ensure that MMC data is not accessed, used, or disclosed for UC Davis Health research purposes.
If protected health information or personal information from the medical records will be stored on an encrypted device, investigators must follow applicable university policies (UC Davis Hospital Policy 1313, UCDHS P&P 2300-2499, and UC Business and Finance Bulletin on Information Security (IS-3)). Please contact the Biomedical Informatics Department for assistance with data security.

If identifiable protected health information is extracted from the UCDH EMR, it may not be re-disclosed/released outside the study team.

UC Davis Student Education Records: If this study involves use of UC Davis students' educational records (including records in the PI's own possession such as course exams/assignments), you must consult with the Registrar's office to see if all requirements of the Family Educational Rights and Privacy Act (FERPA) are satisfied.

10) Risks to Subjects

☐ This research may pose the risk of loss of confidentiality. The risk will be minimized through the processes described above. This study will abide by all applicable law, regulations, and standard operating governing the protection of human subjects, student information and protected health information.

☒ Other – Describe: *This study poses minimal risk to subjects that are no greater than those encountered in daily life.*

11) Potential Benefits to Subjects

☒ Research subjects are not likely to receive any benefit from the proposed research, but others may benefit from the knowledge obtained.

☐ Other – Describe:

12) Sharing Results with Subjects

☒ Results will not be shared with subjects.

☐ Results will be shared with subjects.

a) If results will be shared, describe the results (study results or individual subject results) to be shared with subjects or others (e.g., the subject's primary care physicians):

13) Data Banking

a) Will the data ever be used by you or other researchers to answer a different research aim that is not included in this study?

☒ Yes – Complete the remainder of Section 13.

☐ No – Do not complete Section 13. Go to Section 14.

b) What will be banked for future use?

☒ De-identified data/specimens. Banked data/specimens cannot be linked to an individual.

☐ Identifiable data/specimens – Banked data/specimens will include identifying information.

☐ Coded data/specimens – Banked data/specimens will be stripped of identifiers and assigned a code. A key will be maintained that links the identifiers to the data/specimens.

☐ Contact information will be banked for future research opportunities.

c) Where will the data be banked?

Data will be stored on a secure UC Davis server.

d) How long will the data be banked?

Data will be stored indefinitely, however all personal identifiers will be destroyed.

e) Who will have access to the banked data?

Only the principal investigator and co-investigators.

f) Describe the procedures to release data. Include the process to request a release, approvals required for release, who can obtain data, and the data to be provided:

Data will not be released to the public, and it would only be shared with other researchers upon request. The stored data will not contain any personal identifiers; thus, the sharing of data will not pose any harm for the privacy of subjects.

Note: Identifiable protected health information extracted from the UCDH EMR under an IRB-issued waiver of HIPAA Authorization may not be re-disclosed/released outside the study team.

14) Review Requirement

Some research projects require specific IRB determinations.

a) Are there any contractual obligations or other considerations that require IRB review of this research, or review at intervals other than those required by the Common Rule or FDA?

☐ Yes

☒ No

b) If yes, please describe: